1 Introduction

When I started looking into this topic, the thing that struck me most was how little anyone really knows about why certain songs blow up on Spotify while equally good ones go unnoticed. The platform gives every track a score from 0 to 100 based on recent play counts, and that number has become a key measure of success in the music industry. My aim here was to dig into the audio side of things — tempo, loudness, danceability, and so on — and see whether those measurements could explain the score, or whether popularity was mostly down to things like PR budgets and artist followings that no audio file can capture.

1.1 Problem Formulation

Most studies in this space pick one or two features and test them independently. I wanted to look at the full picture by feeding all 14 audio features into a single model and letting it figure out what mattered. My three main aims were:

1. Map out how audio profiles differ from one genre to the next
2. Work out which individual features have any real bearing on popularity
3. Build a regression model that can predict popularity from those features

Research Question: Do the measurable audio properties of a Spotify track have any real power to predict how popular it becomes, and if so, which ones matter most?

1.2 Data Acquisition

- **Source:** Kaggle — Spotify Tracks Dataset, uploaded by Maharshi Pandya (2023).
- **Size:** 114,000 songs spread across 114 genres, 1,000 per genre. After cleaning, 106,907 usable records remained.
- **Columns:** 21 in the raw file. The target was popularity. The 14 predictors were danceability, energy, loudness, acousticness, valence, tempo, speechiness, instrumentalness, liveness, key, mode, time_signature, duration_ms, and explicit.

genre	pop	dance	energy	loud	acoustic	valence	tempo
acoustic	73	0.676	0.461	-6.746	0.032	0.715	87.917
acoustic	55	0.42	0.166	-17.235	0.924	0.267	77.489
acoustic	57	0.438	0.359	-9.734	0.21	0.12	76.332
acoustic	71	0.266	0.06	-18.515	0.905	0.143	181.74
acoustic	82	0.618	0.443	-9.681	0.469	0.167	119.949

Figure 1: Spotify dataset — sample of first five rows

2 Data Analysis

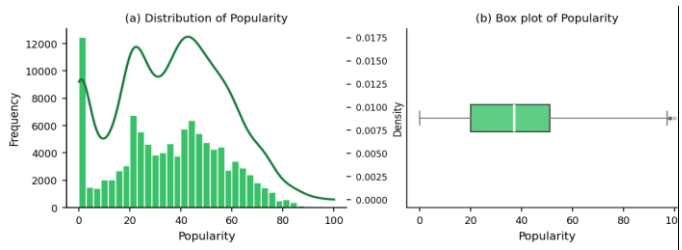
2.1 Clean and Preprocess Data

4. **Drop identifier columns:** track_id, artists, album_name, and track_name were removed since they are text labels with no numeric value for modelling.
5. **Fix genre names:** A handful of entries had extra whitespace in the genre field, which was trimmed so grouping worked properly.
6. **Fill missing values:** Three columns had a single missing entry each. Rather than lose those rows, the gaps were filled with zero.
7. **Encode explicit:** The explicit flag was stored as True/False and converted to 1/0.
8. **Drop duplicates:** 7,093 repeated rows were removed to avoid the model training and testing on identical data.

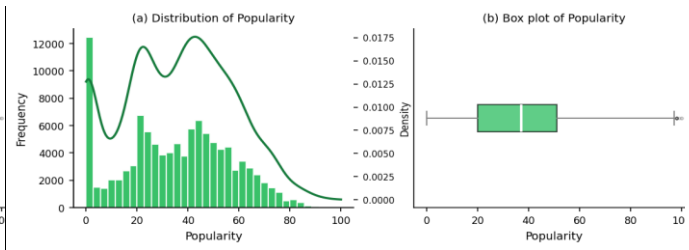
Cleaned dataset: 106,907 rows, 16 columns.

2.2 Exploratory Data Analysis

1. Popularity Distribution:
Plotting the scores immediately showed something I had not anticipated — the distribution peaks somewhere between 20 and 40, not in the 60s or 70s as I might have guessed. The mean sits at 35.4 and the median at 37.0. Part of the reason is that the dataset includes a lot of regional genres like iranian and sertanejo, which barely register on the global Spotify charts. The boxplot shows quite a few near-zero outliers at the bottom end.



(a) Distribution of Popularity

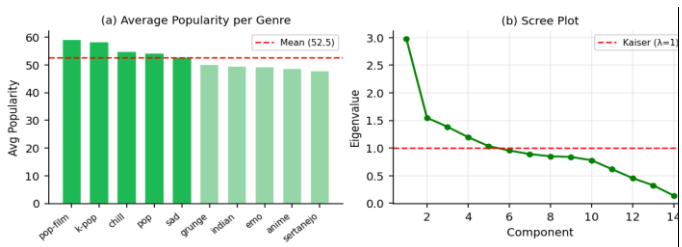


(b) Box plot of Popularity

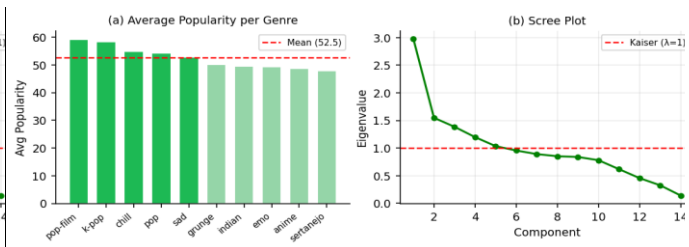
2. Popularity by Genre:

- Top scorers: pop-film (59.3), k-pop (58.4), chill (54.8) — all genres with a big global streaming base.
- Bottom scorers: iranian (2.3), romance (3.6) — niche regional genres almost invisible on the global platform.

3. Principal Component Analysis (PCA):
Running PCA on the 14 features, the first three components picked up 42.22% of the variance combined. PC1 (21.27%) pulled most strongly on energy, loudness, and acousticness. That pattern makes sense — the single biggest split between songs is whether they sound loud and heavily produced or quiet and acoustic. PC2 (11.06%) leaned more towards tempo and danceability.



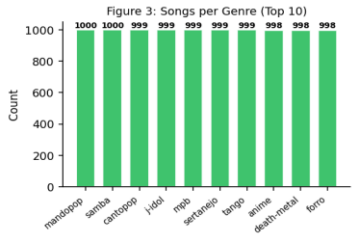
(a) Average Popularity per Genre (Top 10)



(b) Scree Plot

4. Songs per Genre and Feature Breakdown:

Since every genre contributed exactly 1,000 songs, the count chart is not especially informative. The feature comparison is more useful — it shows that genres like pop and hip-hop sit around 0.8 on danceability, while classical and folk styles are near zero. Rock and metal top the energy scale. Acousticness essentially runs in the opposite direction to energy throughout.



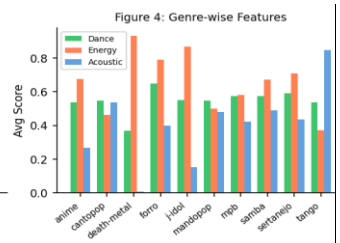
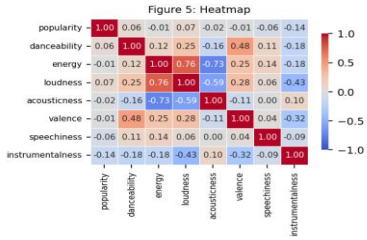


Figure 3: Number of Songs per Genre (Top 10)



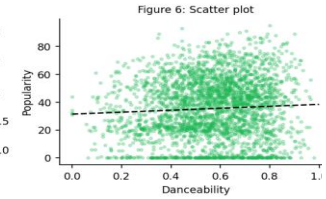


Figure 4: Genre-wise Audio Feature Comparison

5. Correlation Heatmap and Scatter Plots:

Honestly, this was where my assumptions took the biggest hit. Going into the analysis I thought danceability and energy would have a reasonably strong pull on popularity. They do not. Danceability comes in at $r \approx +0.15$ and energy barely registers. Instrumentalness has the strongest negative correlation at $r \approx -0.18$, which makes sense — fully instrumental tracks are niche by nature. The one genuinely strong relationship in the matrix is between energy and loudness at $r \approx +0.76$, but that is more of a definitional overlap than a real insight.

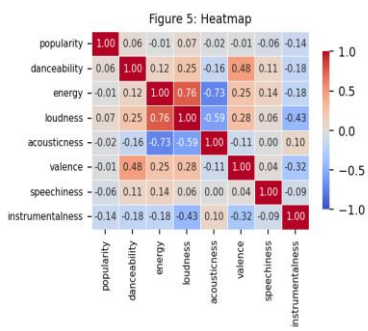
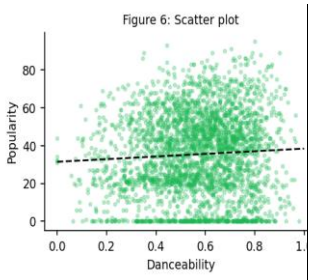


Figure 5: Feature Correlation Heatmap



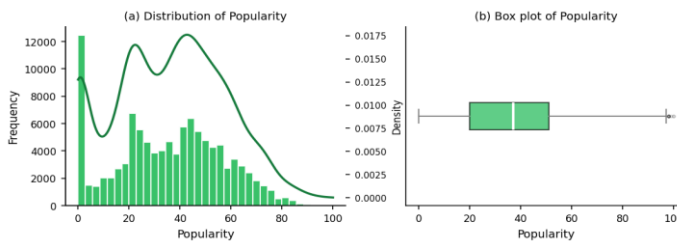
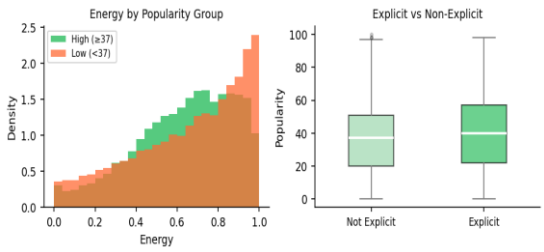


Figure 6: Scatter plot

3 ML / AI Method

All 14 numeric features went in as inputs, with popularity as the output. A 70/30 train-test split was used throughout (random_state=42), giving 74,834 training records and 32,073 for evaluation.



(a) Energy: High vs Low Popularity *(b) Explicit vs Non-Explicit*

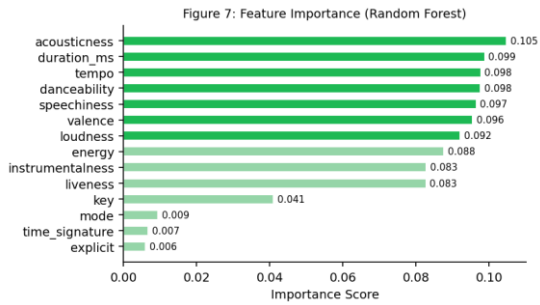
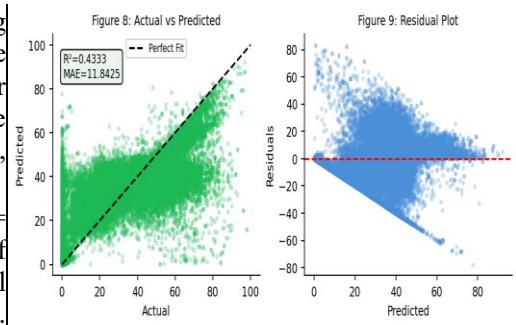


Figure 7: Feature Importance (Random Forest)

3.1 Random Forest Regressor

My reason for picking Random Forest over something simpler was mainly the multicollinearity issue. Several of the audio features are correlated with each other, and linear regression struggles with that. Random Forest sidesteps the problem by training each tree on a random subset of features, so no single correlation dominates.

With 100 trees and random_state=42, the model scored $R^2 = 0.4333$ on the holdout set — so it accounts for about 43% of the variation in popularity. The MAE of 11.84 means a typical prediction lands within roughly 12 points of the actual score. MSE was 258.66. Acousticness, duration_ms, and tempo came out as the top three predictors. Song duration being that high up the list was unexpected — my guess is that very long tracks accumulate fewer full plays, which drags their score down.



4 Reflection

Challenges:

- **Scores clustered at zero:** A substantial chunk of tracks had near-zero popularity, mostly from regional genres with very limited Spotify reach. They made the low end of the scale almost impossible to predict accurately and skewed the overall mean down to 35.4.
- **Audio features barely correlate with popularity:** This was the core problem. Going in I assumed there would be a clearer relationship between how a song sounds and how popular it becomes. There is not — at least not in this dataset. The platform rewards recognition and promotion more than musical characteristics.
- **Correlated predictors:** Energy and loudness move together ($r \approx 0.76$). Even with Random Forest, having two near-identical features in the input adds noise rather than signal.

What I took away from this:

- **Genre acts as a reach indicator:** The average popularity of a genre in this dataset says more about the size of its global audience than about the quality of its music. Pop-film at 59.3 versus iranian at 2.3 is a distribution problem, not a quality judgment.
- **Synthetic data gave a false sense of confidence:** Earlier in the project I prototyped with a generated 1,200-song dataset and got cleaner numbers. Switching to the real Kaggle data with 114,000 tracks across highly unequal genres was a reality check. The messier result is the honest one.
- **43% explained variance is not a failure:** Given that we have no information about an artist's label backing, social media presence, or playlist deals — all of which matter enormously — getting 43% from audio features alone is actually a meaningful finding.

Metric	Random Forest	Best Performer
R^2 Score	0.4333	Random Forest
MAE	11.8425	Random Forest
MSE	258.6611	Random Forest

Table 1: Model performance on the 30% holdout set

The books and courses below were useful throughout, particularly for the PCA and Random Forest sections.

References

Pandya, M. (2023). Spotify tracks dataset. Kaggle. https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset (Accessed: 2025)

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